STRUCTURAL WEAKNESSES ALLOW THE ABC TRANSPORTER TO DEFORM WITHIN THE MEMBRANE

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ABSTRACT

ATP-binding cassette (ABC) transporters constitute one of the largest families of membrane proteins and play a key role in cellular detoxification, lipid trafficking, and multidrug resistance. These proteins must undergo conformational deformations to translocate substrates across the hydrophobic barrier of the lipid bilayer. Yet, at the same time, they are constrained by the membrane that exerts a lateral pressure on their structure. Structural evidence suggests that they undergo deformations using helical discontinuities in trans-membrane segments acting as mechanical joints allowing them to deform and change conformations. This observation is also observed in the conformational landscape derived from inter-particle variability of cryoEM data, showing helical deformations stemming from these kinks.

We undertook a statistical analysis of the PDB to understand how these deformations distribute compared to the membrane bilayer for the type IV family of ABC transporters (Figure). We could visualize how the kinks distribute in each trans-membrane helix, and well as their degree of deformation. We also investigated the distribution of kinks in different conformations sampled by these ABC transporters (from Inward-to outward-facing). Altogether, this analysis brings a statistical view of how a whole family of transporter uses fragility within their framework to perform conformational changes.

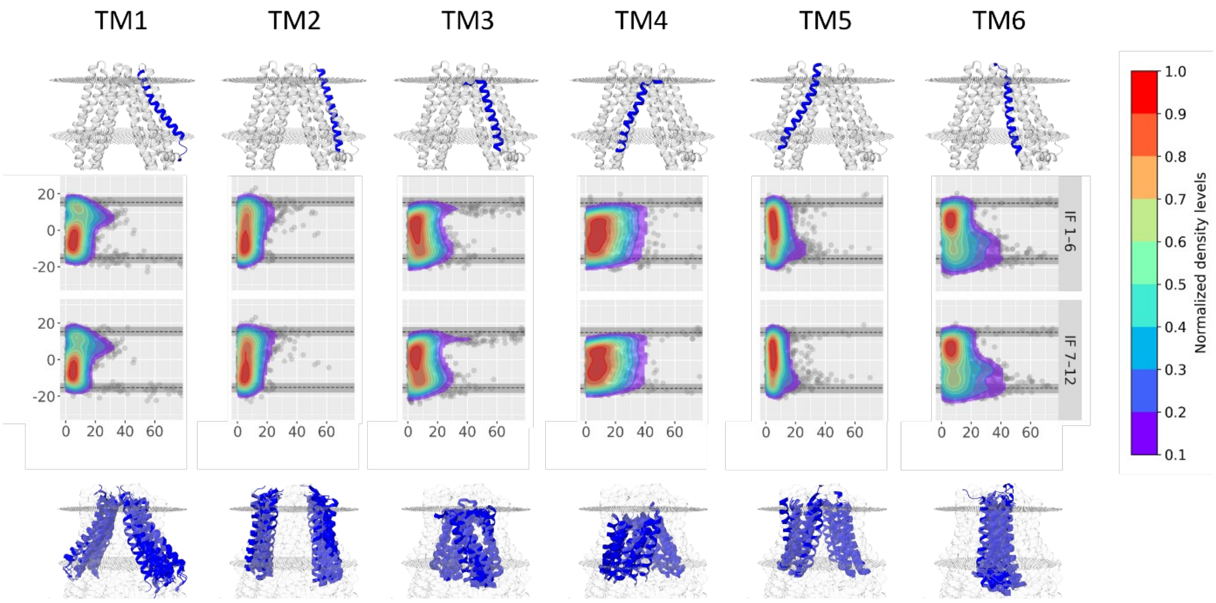

Fig. *Distribution of TMD structural deformation in the type IV family of homodimer ABC transporters for the IF conformation*